CLASS 10 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 5: **Arithmetic Progressions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $a_n = 3n + 1$, then the common difference is: 1

(a) 1 **(b)** 6 **(c)** 3 **(d)** 4

2. If $S_n = n^2 + 5n$, then $a_1 =$ 1

(a) 6 **(b)** 7 **(c)** 5 **(d)** 1

3. The sum of first 12 positive odd numbers is: 1

(a) 132 **(b)** 288 **(c)** 156 **(d)** 144

4. The 10th term of the AP $2, \frac{7}{2}, 5, \frac{13}{2}, \ldots$ is: 1

(a) 15 **(b)** $\frac{31}{2}$ **(c)** 17 **(d)** 14

5. The 7th term of $11, 7, 3, -1, \ldots$ is: 1

(a) -9 **(b)** -28 **(c)** -13 **(d)** -17

6. The 10th term from the end of the AP with last term 60 and $d = 2$ is: 1

(a) 42 **(b)** 44 **(c)** 78 **(d)** 40

7. If $a_n = 3n + 9$, then the common difference is: 1

(a) 9 **(b)** 12 **(c)** 3 **(d)** 6

8. If $S_n = n^2 + 6n$, then $a_1 =$ 1

(a) 1 **(b)** 6 **(c)** 7 **(d)** 8

9. The sum of first 15 positive odd numbers is: 1

(a) 225 **(b)** 240 **(c)** 450 **(d)** 210

10. The 10th term of the AP $9, 11, 13, 15, \ldots$ is: 1

(a) 29 **(b)** 27 **(c)** 25 **(d)** 20

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the sum of the AP $12, 10, \ldots, -6$. 2

12. Which term of the AP $20, 17, 14, 11, \ldots$ is -1? 2

13. Find the sum of the first 20 multiples of 7. 2

14. Find the sum of the AP $2, 7, \ldots, 97$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In an AP, the 3rd term is 11 and the 8th term is 36. Find the AP and its 11th term. 3

16. How many terms are there in the AP $4, 10, 16, \ldots, 106$? 3

17. Find the sum of the first 18 terms of the AP $5, 13, 21, 29, \ldots$ 3

OR

In an AP, the 2nd term is 9 and the 10th term is 25. Find the AP and its 20th term. 3

18. How many terms are there in the AP $12, 17, 22, \ldots, 117$? 3

OR

Find the sum of the first 16 terms of the AP $21, 27, 33, 39, \ldots$ 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. How many terms of the AP $17, 23, 29, 35, \ldots$ must be taken so that the sum is 517? 5

OR

A loan is repaid in 15 monthly instalments; the first is ₹300 and each later instalment is ₹20 more than the previous. Find the instalment in the last month and the total amount repaid. 5

20. How many terms of the AP $45, 41, 37, 33, \ldots$ must be taken so that the sum is 270? 5

OR

A loan is repaid in 8 monthly instalments; the first is ₹1000 and each later instalment is ₹50 more than the previous. Find the instalment in the last month and the total amount repaid. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND